SIMoN

SIMoN says go home!

LUNAR LAVA TUBE ABSOLUTE POSITIONING

Mission Summary

ACCURACY	2 – 5 m 3D position inside lava tubes
DEPTH RANGE	Up to 100 m underground operational depth
SETUP TIME	< 24 h from survey to operational use
PAYLOAD (ROVER)	~ 150 kg mapping & navigation equipment

MISSION CONCEPT

SIMoN enables absolute positioning inside lunar lava tubes by extending surface-based navigation into the subsurface. The system combines lunar satellite navigation, LiDAR-based mapping, and ground-penetrating radar to create a georeferenced 3D model that allows astronauts to navigate safely and accurately underground.

SYSTEM OVERVIEW

SURFACE NAVIGATION	SUBSURFACE MAPPING	SURFACE GPR SCAN	DATA PROCESSING	USER APPLICATION
GNSS Navigation Signals	LiDAR-IMU Survey	Detect Buried Markers	Georeferencing & 3D Model	Real-time Navigation

Absolute surface position → High-resolution point cloud → Marker detection & depth estimation → Georeferenced 3D model → Continuous absolute positioning

MISSION IMPACT

- Enables safe human exploration in dark, GNSS-denied environments.
- Supports high-value scientific operations and resource prospecting.
- Lays the foundation for future subsurface habitats and infrastructure.
- Transforms unmapped voids into precisely navigable spaces.

BUSINESS MODEL

HARDWARE SALES
Rovers, LiDAR-IMU, GPR systems, markers

SOFTWARE LICENSING
Mapping & positioning software platform

SERVICE CONTRACTS
Data processing, model updates, mission support

Infrastructure-based model generating recurring revenue through long-term service and software subscriptions.

MARKET POTENTIAL & PROJECTIONS

	2030	2035	2040	2045+
Deployed Systems (Cumulative)	1–2	5	10	20+
Revenue (Cumulative)	~ €5M	~ €30M	~ €90M	~ €250M+
Gross Profit Margin	35%	40%	45%	50%+
Market Impact Stage	Technology Demonstration	Early Adoption	Infrastructure Expansion	Operational Standard

Projections reflect a conservative outlook for an emerging high-value market.

LUNAR INFRASTRUCTURE TIMELINE (REFERENCE)

2027–2029	2030	2032–2035	2038	2040+
Precursor Missions Technology demos, payload tests, lunar scouting	**MOONLIGHT FOC** Moonlight GNSS constellation fully operational in lunar orbit	**Permanent Lunar Infrastructure** Establish surface infrastructure & habitats	**First Lava Tube Exploration Missions** Robotic & human exploration of subsurface	**SIMoN Operational Deployment** Operational use of SIMoN for subsurface navigation

SIMoN builds upon the Moonlight GNSS infrastructure and extends positioning capabilities into the lunar subsurface.

LONG-TERM OUTLOOK

As lunar exploration shifts from flag-and-footprints to permanent presence, subsurface navigation becomes a critical infrastructure.

SIMoN is positioned to become the de-facto standard for lunar lava tube operations, enabling safe return and expanding humanity's reach into the Moon's hidden frontier.

OUR TEAM

Michael Nacke · Michal Pushkov · Andrea Tantucci · Dario Blandi · Karolina Sobecka · Mutaz Qafisheh · Zubair Akhter · Hugues Ruisseau · Suhui Jeong

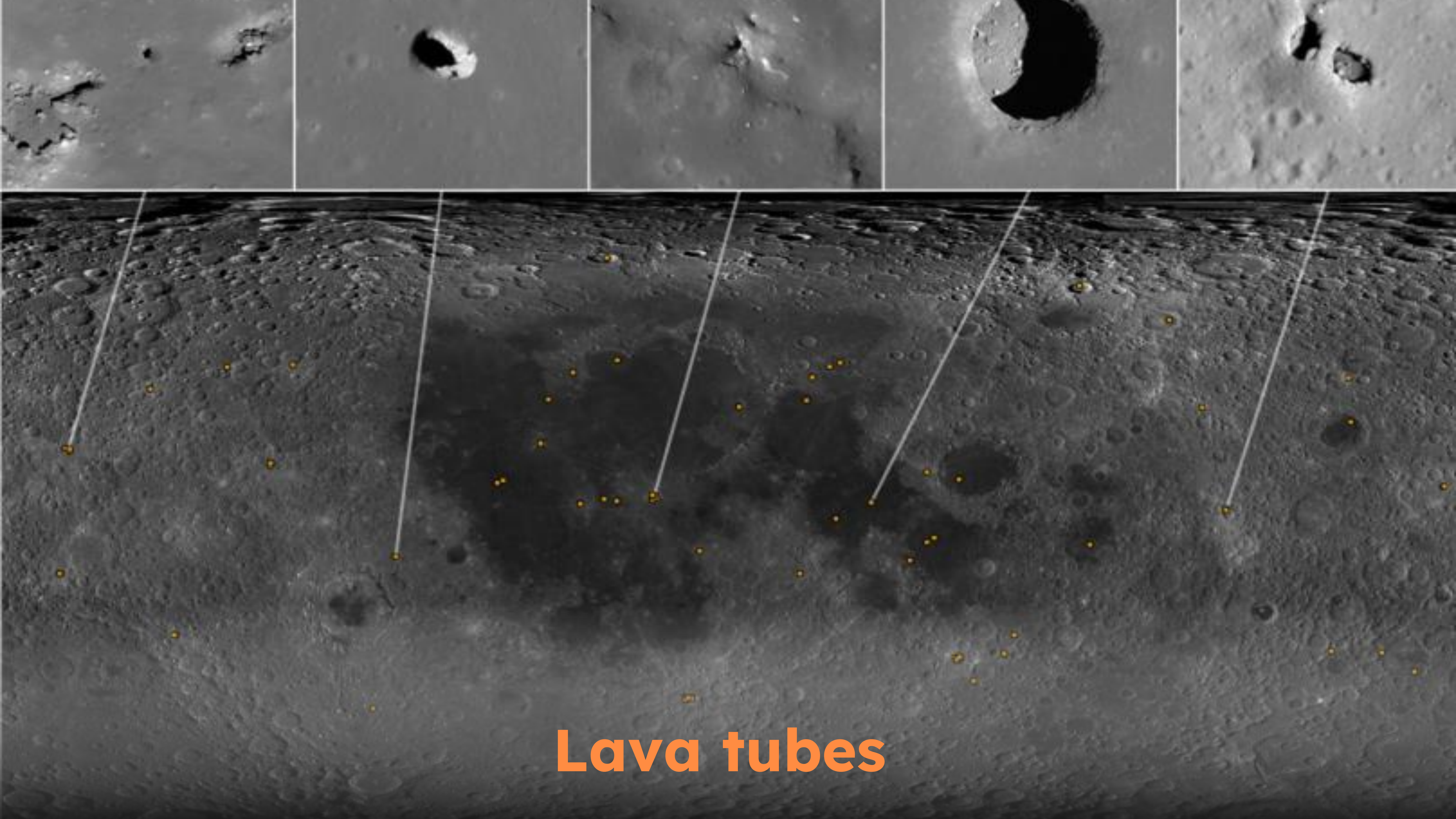
EXPLORING FURTHER. NAVIGATING SMARTER. COMING HOME.

PROJECT
SIMoN

by Navola Team

A hybrid positioning system for safe, reliable navigation inside lunar lava tubes: the first underground localisation ecosystem beyond Earth.

ESA / JRC GNSS Summer School 2026
Group Project Proposal | Final Presentation, 23 July 2026


Lava tubes

Nature already built the shelter

-95%

RADIATION

Lava tubes could reduce radiation exposure from ~100–400 mSv/year on the surface to just a few mSv/year underground.

-25°C

STABLE INTERIOR

Surface temperatures can vary from approximately +120°C in sunlight to below −150°C in darkness.

<1%

ESTIMATED IMPACT RISK

The rock ceiling shields against micrometeorites and the ejecta they throw off.

100+ m

WIDE, 10 – >100 km KM LONG

Tubes such as Lacus Mortis offer room for a full base, not just a shelter.

Sources: Kalita et al. 2021; Naito et al.; NASA, ESA, SETI Institute; Lunar Vertex 2026.

PROBLEMS

No positioning exists below the lunar surface

Habitats go underground

Future lunar bases will be sited inside lava tubes for natural protection, away from any surface infrastructure.

No GNSS reaches inside

Satellite signals cannot penetrate metres of overlying basalt, so surface navigation stops at the tube entrance.

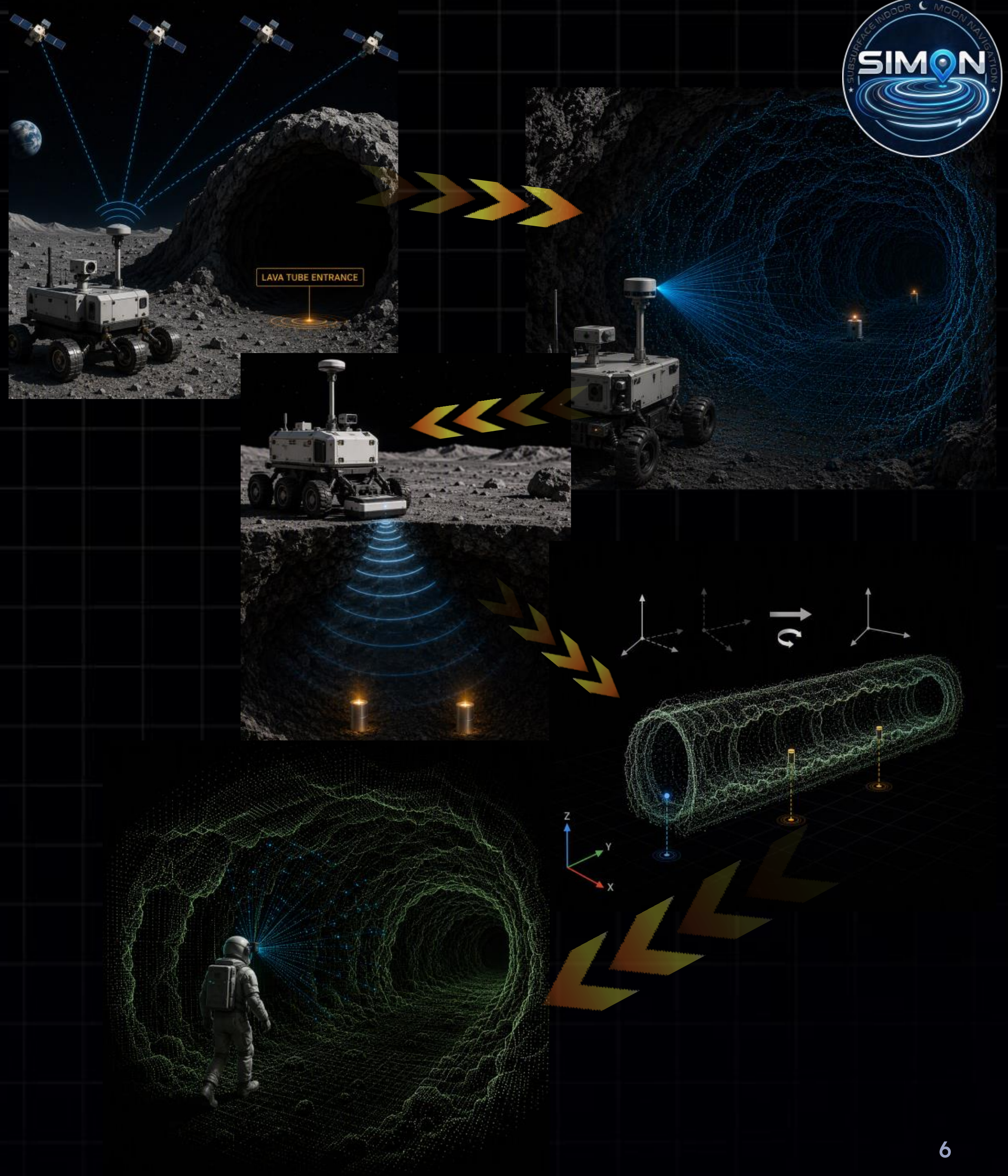
Crews and robots need to navigate

Astronauts and autonomous systems require continuous, reliable positioning to operate and stay safe.

SOLUTION

Sub-surface navigation

1. The position of the lava tube entrance is determined with the Moonlight navigation system.
2. Entering the tube, the rover builds a high-resolution 3D point cloud (LiDAR+IMU) and deploys metallic control markers.
3. GPR detects the buried markers, while Moonlight provides the surface coordinates above them. Marker positions are then derived using GPR depth measurements.
4. The entrance and control points are used to align the tunnel model within the lunar reference system.
5. Continuous absolute positioning is achieved through wearable LiDAR-IMU measurements matching against the referenced 3D lava tube model.



Sources: Calvo et al., 2026; Chunyang et al., 2025; Wen et al., 2023; Qiao et al., 2015.

USE CASES

One infrastructure, many use cases

Crew Wayfinding

Astronaut navigation underground.

Search & Rescue

Emergency crew localization.

Multi-Robot Fleets

Coordinated rover operations.

Asset Tracking

Equipment and infrastructure tracking.

Building operations

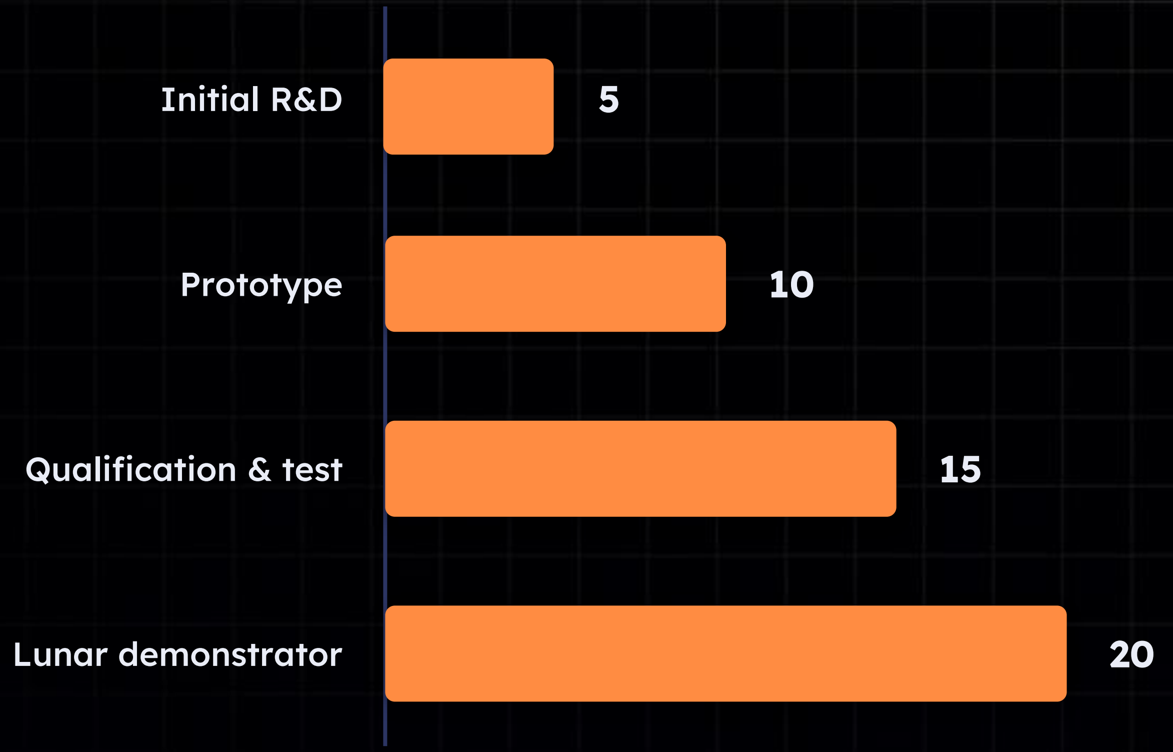
Infrastructure building and interface with the surface.

Beyond the Moon

Scalable to Martian lava tubes.

Phased investment, recurring returns

Development capex by phase (€ M)



Revenue model

- Navigation hardware deployment
- Maintenance and support contracts
- Infrastructure-as-a-Service (IaaS)
- Long-term government and commercial partnerships
- Navigation hardware deployment

€10 M / yr target recurring revenue from annual service contracts at scale

A small, high-value strategic market

Institutional

ESA	Moonlight & lunar exploration
NASA	Artemis programme
LUPEX (JAXA + ISRO)	Polar exploration
CSA	Partner missions

Industry

SpaceX	Lunar operations
Blue Origin	Moon base
Mining companies	In-Situ Resource Utilization (ISRU)
Rover manufacturers	Autonomous systems

Market signal: In spring 2026 NASA raised the cumulative ceiling of its Commercial Lunar Payload Services (CLPS) contract from **$2.6 B to $4.2 B**, a 60% jump that signals rapidly expanding public investment in lunar delivery.

From demonstration to lunar economy

2030
Demonstration Mission

Prototype, qualification, and a lunar demonstrator.

2032-35
Initial Infrastructure

First operational deployments at pioneer sites.

2040-45
Permanent Habitats

Scaled service across habitats and mining at lunar sites.

>2045
Beyond the Moon

Extension to Mars.

INNOVATION

What sets SIMoN apart

	Terrestrial mining Syntony, Advanced Nav	Lunar rovers NASA LUX, ESA Skylight, SWIPE, TruewaveGPR	SIMoN
Works underground (GNSS-denied)	✓	✓	✓
Operates on the Moon	✗	✓	✓
Absolute lunar georeferencing	✗	✗	✓
Multi-user shared ecosystem	✗	✗	✓
Built for lunar lava tubes	✗	✓	✓

The gap: mining systems position underground but not on the Moon; lunar rovers work on the surface but not below it. SIMoN is the only system built for both.

CONCLUSION

Technically feasible

Integrates flight-proven LiDAR, IMU, GPR, and Moon PNT.

Financially sound

Phased capex, high-value contracts, recurring revenue.

Culturally aligned

Enhances safety and cooperation; no privacy concerns.


Enabling humanity's next step beneath the lunar surface.

THANKS FOR THE ATTENTION

NAVOLA Team

- Michal Pushkov - Project Manager
- Andrea Tantucci - Product Manager
- Zubair Akhter - Technical Team
- Hugues Ruisseau - Technical Team
- Karolina Sobecka - Design Inspector
- Dario Blandi - Creative Director
- Micheal Nacke - Business Responsible
- Suhui Jeong - VFX Designer
- Mutaz Qafisheh - Technical Team

SIMoN says...